Python Programming

Case-Based Examples: Data Types and Operators

Each case below describes a small real-world scenario. Read the scenario, study the solution code, and note which data types and operators are used. Try modifying the input values and predict the new output before running the code.

Case 1: Movie Ticket Booking

Scenario:

A cinema charges ■250 per ticket. A customer wants to book 4 tickets and pay using a coupon that gives a flat ■100 discount if the total bill is above ■500. Calculate the final amount payable

Concepts used: int, Arithmetic operators (*), Comparison operator (>), Assignment operator (-=)

Case 2: Auto Username Generator

Scenario:

A college wants to auto-generate a login username for every student using their first name and the last two digits of their roll number, in lowercase, with no spaces

Concepts used: str, String methods (lower(), slicing), String concatenation (+)

Case 3: ATM Cash Withdrawal

Scenario:

An ATM allows withdrawal only if the requested amount does not exceed the account balance AND the amount is a multiple of 100. Check whether a withdrawal request of ■4500 from a balance of ■10000 is valid.

Concepts used: int, bool, Comparison operators (<=, ==), Logical operator (and), Arithmetic (%)

Case 4: Employee Salary and Bonus

Scenario:

An employee's base salary increases by a 10% bonus every year for 3 consecutive years. Track and print the salary after each year using an assignment operator

Concepts used: int/float, Assignment operator (+=), Arithmetic operator (*)

Case 5: Traffic Signal Validator

Scenario:

A traffic simulation program needs to check whether a given signal color entered by a user is a valid traffic light color (red, yellow, green).

Concepts used: str, list, Membership operator (in)

Case 6: Duplicate Student Record Check

Scenario:

A school database wants to verify whether two student record variables actually point to the same record in memory, or whether they are just two separate copies with the same data.

Concepts used: dict, Identity operators (is), Equality operator (==)

Case 7: File Permission Flags

Scenario:

An operating system stores file permissions as bit flags: Read = 4, Write = 2, Execute = 1. A file has Read and Write permission. Combine the flags and later check if the Write permission is set.

Concepts used: int, bool, Bitwise operators (|, &)

Case 8: Scholarship Eligibility Check

A student is eligible for a merit scholarship if their percentage is above 85, OR (their percentage is above 75 AND their family income is below ■2,00,000).

Concepts used: int, bool, Logical operators (and, or), Comparison operators (>, <)